Laboratory Work 2 - End-to-End Data Analytics in Power BI: From Data Cleaning to DAX Insights

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IS 107 - Business Intelligence

Part 1: Open Power Query Editor

Untitled - Power BI Desktop

Untitled - Power Query Editor

= Table.TransformColumnTypes(#"Promoted Headers",{{"OrderID", type text},

	OrderID	OrderDate	CustomerName	Product
1	ORD-346	2025-01-19	Mia Tan	Monitor
2	ORD-123	2025-02-05	Paul Yu	Keyboard
3	ORD-379	08-02-2025	Pedro Reyes	Laptop
4	ORD-132	2025/02/27	Chris Go	Headset
5	ORD-016	13-01-2025	Ana Lopez	Webcam
6	ORD-213	2025-02-22	Bea Lim	Monitor
7	ORD-161	02/23/2025	Carlo Diaz	Headset
8	ORD-093	2025-02-08	MARIA SANTOS	Webcam
9	ORD-254	2025-01-24	Ella Yu	Mouse
10	ORD-404	2025-02-28	Ella Yu	Laptop
11	ORD-225	Feb 12 2025	Mia Tan	Monitor
12	ORD-360	2025-01-28	MARIA SANTOS	Webcam
13	ORD-050	Feb 13 2025	Chris Go	Headset
14	ORD-183	01-02-2025	Bea Lim	Mouse

7 COLUMNS, 500 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 11:38 PM

Queries [1]: Messy_Sales_Data xlsx -...

Query Settings

PROPERTIES

Name: Messy_Sales_Data xlsx - MessySales

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Page 1 of 1

Part 2: Remove Duplicates

```
= Table.Distinct(#"Changed Type", {"OrderID", "OrderDate"})
```

	OrderID	OrderDate	CustomerName	Product	Quantity	SalesAmoun
1	ORD-346	2025-01-19	Mia Tan	Monitor	2	7500
2	ORD-123	2025-02-05	Paul Yu	Keyboard	1	45000
3	ORD-379	08-02-2025	Pedro Reyes	Laptop	2	3000
4	ORD-132	2025/02/27	Chris Go	Headset	1	7500
5	ORD-016	13-01-2025	Ana Lopez	Webcam	1	7500
6	ORD-213	2025-02-22	Bea Lim	Monitor	null	1500
7	ORD-161	02/23/2025	Carlo Diaz	Headset	3	"7500"
8	ORD-093	2025-02-08	MARIA SANTOS	Webcam	2	800
9	ORD-254	2025-01-24	Ella Yu	Mouse	1	3000
10	ORD-404	2025-02-28	Ella Yu	Laptop	3	2200
11	ORD-225	Feb 12 2025	Mia Tan	Monitor	3	3000
12	ORD-360	2025-01-28	MARIA SANTOS	Webcam	3	45000
13	ORD-050	Feb 13 2025	Chris Go	Headset	2	1200
14	ORD-183	01-02-2025	Bea Lim	Mouse	3	7500
15	ORD-316	26-01-2025	Liza Ramos	Monitor	null	1500
16	ORD-310	2025-02-14	Mia Tan	Headset	null	2200
17	ORD-082	Feb 19 2025	Chris Go	Monitor	2	3000
18	ORD-404	07-02-2025	Sarah Ong	Laptop	3	1200
19	ORD-328	04-01-2025	Rico Tan	Headset	1	3000
20	ORD-133	02/12/2025	Rico Tan	Keyboard	1	"7500"
21	ORD-271	Jan 31 2025	Mark Tan	Headset	3	7500
22	ORD-252	2025/02/14	Tom Lee	Laptop	2	1200
23	ORD-373	2025/01/11	Mark Tan	Keyboard	1	"7500"
24	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Query Settings

PROPERTIES

Name

Messy_Sales_Data.xlsx - MessySales

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Removed Duplicates

profiling based on top 1000 rows

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Part 3: Handle Missing Values

```
= Table.FillDown(#"Removed Blank Rows",{"OrderID", "OrderDate"})
```

	OrderID	OrderDate	CustomerName	Product	Quantity	SalesAmou
1	0	2025-01-19	Mia Tan	Monitor	2	7500
2	ORD-123	2025-02-05	Paul Yu	Keyboard	1	45000
3	ORD-379	08-02-2025	Pedro Reyes	Laptop	2	3000
4	ORD-132	2025/02/27	Chris Go	Headset	1	7500
5	ORD-016	13-01-2025	Ana Lopez	Webcam	1	7500
6	ORD-213	2025-02-22	Bea Lim	Monitor	null	1500
7	ORD-161	02/23/2025	Carlo Diaz	Headset	3	"7500"
8	ORD-093	2025-02-08	MARIA SANTOS	Webcam	2	800
9	ORD-254	2025-01-24	Ella Yu	Mouse	1	3000
10	ORD-404	2025-02-28	Ella Yu	Laptop	3	2200
11	ORD-225	Feb 12 2025	Mia Tan	Monitor	3	3000
12	ORD-360	2025-01-28	MARIA SANTOS	Webcam	3	45000
13	ORD-050	Feb 13 2025	Chris Go	Headset	2	1200
14	ORD-183	01-02-2025	Bea Lim	Mouse	3	7500
15	ORD-316	26-01-2025	Liza Ramos	Monitor	null	1500
16	ORD-310	2025-02-14	Mia Tan	Headset	null	2200
17	ORD-082	Feb 19 2025	Chris Go	Monitor	2	3000
18	ORD-404	07-02-2025	Sarah Ong	Laptop	3	1200
19	ORD-328	04-01-2025	Rico Tan	Headset	1	3000
20	ORD-133	02/12/2025	Rico Tan	Keyboard	1	"7500"
21	ORD-271	Jan 31 2025	Mark Tan	Headset	3	7500
22	ORD-252	2025/02/14	Tom Lee	Laptop	2	1200
23	ORD-373	2025/01/11	Mark Tan	Keyboard	1	"7500"
24	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Query Settings

PROPERTIES

Name

Messy_Sales_Data xlsx - MessySales

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Removed Duplicates
- Replaced Value
- Removed Blank Rows
- Filled Down

rofiling based on top 1000 rows

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Part 4: Change Data Types

```
= Table.RemoveRowsWithErrors(#"Removed Errors", {"SalesAmount"})
```

	OrderDate	CustomerName	Product	Quantity	SalesAmount	Regio
1	1/19/2025	Mia Tan	Monitor	2	7500	Luzon
2	2/5/2025	Paul Yu	Keyboard	1	45000	
3	8/2/2025	Pedro Reyes	Laptop	2	3000	Luzon
4	2/27/2025	Chris Go	Headset	1	7500	Minda
5	2/22/2025	Bea Lim	Monitor	null	1500	Luzon
6	2/8/2025	MARIA SANTOS	Webcam	2	800	Luzon
7	1/24/2025	Ella Yu	Mouse	1	3000	Minda
8	2/28/2025	Ella Yu	Laptop	3	2200	Visaya
9	2/12/2025	Mia Tan	Monitor	3	3000	Minda
10	1/28/2025	MARIA SANTOS	Webcam	3	45000	Visaya
11	2/13/2025	Chris Go	Headset	2	1200	
12	1/2/2025	Bea Lim	Mouse	3	7500	Visaya
13	2/14/2025	Mia Tan	Headset	null	2200	Luzon
14	2/19/2025	Chris Go	Monitor	2	3000	Minda
15	7/2/2025	Sarah Ong	Laptop	3	1200	
16	4/1/2025	Rico Tan	Headset	1	3000	
17	1/31/2025	Mark Tan	Headset	3	7500	Visaya
18	2/14/2025	Tom Lee	Laptop	2	1200	Visaya
19	2/11/2025	Chris Go	Keyboard	null	1500	
20	2/10/2025	Lea Cruz	Headset	2	2200	Luzon
21	1/28/2025	Chris Go	Mouse	1	800	Visaya
22	1/23/2025	Sarah Ong	Webcam	2	3000	
23	1/1/2025	Nina Ong	Webcam	null	1500	Visaya
24	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Query Settings

PROPERTIES

Name

Messy_Sales_Data.xlsx - MessySales

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Removed Duplicates
- Replaced Value
- Removed Blank Rows
- Filled Down
- Changed Type1
- Removed Errors
- Removed Errors1

Part 5: Rename Columns Properly

```
= Table.RenameColumns(#"Removed Errors1",{{"OrderID", "order_id"}, {"OrderDate", "order_date"}, {"CustomerName",
```

	order_id	order_date	customer_name	Product	Quantity	total_sales
1	0	1/19/2025	Mia Tan	Monitor	2	
2	ORD-123	2/5/2025	Paul Yu	Keyboard	1	
3	ORD-379	8/2/2025	Pedro Reyes	Laptop	2	
4	ORD-132	2/27/2025	Chris Go	Headset	1	
5	ORD-213	2/22/2025	Bea Lim	Monitor	null	
6	ORD-093	2/8/2025	MARIA SANTOS	Webcam	2	
7	ORD-254	1/24/2025	Ella Yu	Mouse	1	
8	ORD-404	2/28/2025	Ella Yu	Laptop	3	
9	ORD-225	2/12/2025	Mia Tan	Monitor	3	
10	ORD-360	1/28/2025	MARIA SANTOS	Webcam	3	
11	ORD-050	2/13/2025	Chris Go	Headset	2	
12	ORD-183	1/2/2025	Bea Lim	Mouse	3	
13	ORD-310	2/14/2025	Mia Tan	Headset	null	
14	ORD-082	2/19/2025	Chris Go	Monitor	2	
15	ORD-404	7/2/2025	Sarah Ong	Laptop	3	
16	ORD-328	4/1/2025	Rico Tan	Headset	1	
17	ORD-271	1/31/2025	Mark Tan	Headset	3	
18	ORD-252	2/14/2025	Tom Lee	Laptop	2	

3

Query Settings

PROPERTIES

Name

Messy_Sales_Data.xlsx - MessySales

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Removed Duplicates
- Replaced Value
- Removed Blank Rows
- Filled Down
- Changed Type1
- Removed Errors
- Removed Errors1
- Renamed Columns

Guide Questions

1. Why is removing duplicates important in data analysis?

   - Duplicates cause inflated totals and inaccurate counts, leading to unreliable reports and wrong business decisions.

2. What problems can missing values cause in reports?

   - Missing values can cause calculation errors, incomplete visuals, and misleading results in reports.

3. How does correct data typing affect calculations?

   - Wrong data types prevent calculations from working — for example, numbers stored as text can't be summed or averaged.

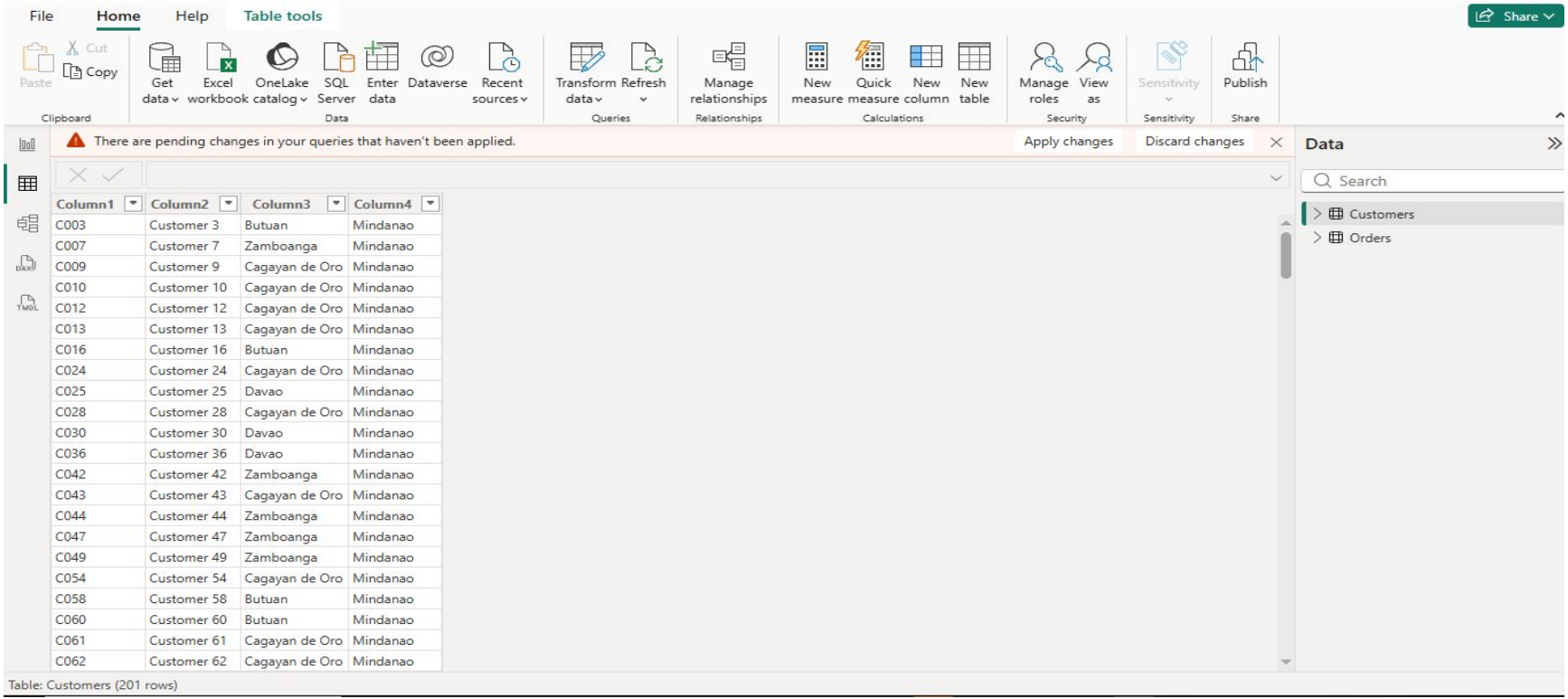
4. What naming conventions improve data readability?

   - Consistent, lowercase names like order_id make columns easier to read, reference, and understand across the team.

Enhancement Activities

- Create a data quality checklist for future datasets.
- Compare dataset size before and after cleaning.
- Research: What is ETL and how does Power Query support it?

Laboratory Activities
Part 1: Load Data



Column1	Column2	Column3	Column4
C003	Customer 3	Butuan	Mindanao
C007	Customer 7	Zamboanga	Mindanao
C009	Customer 9	Cagayan de Oro	Mindanao
C010	Customer 10	Cagayan de Oro	Mindanao
C012	Customer 12	Cagayan de Oro	Mindanao
C013	Customer 13	Cagayan de Oro	Mindanao
C016	Customer 16	Butuan	Mindanao
C024	Customer 24	Cagayan de Oro	Mindanao
C025	Customer 25	Davao	Mindanao
C028	Customer 28	Cagayan de Oro	Mindanao
C030	Customer 30	Davao	Mindanao
C036	Customer 36	Davao	Mindanao
C042	Customer 42	Zamboanga	Mindanao
C043	Customer 43	Cagayan de Oro	Mindanao
C044	Customer 44	Zamboanga	Mindanao
C047	Customer 47	Zamboanga	Mindanao
C049	Customer 49	Zamboanga	Mindanao
C054	Customer 54	Cagayan de Oro	Mindanao
C058	Customer 58	Butuan	Mindanao
C060	Customer 60	Butuan	Mindanao
C061	Customer 61	Cagayan de Oro	Mindanao
C062	Customer 62	Cagayan de Oro	Mindanao

Part 2: Merge Datasets (Inner Join)

```
= Table.ExpandTableColumn(#"Merged Queries", "Customers", {"CustomerID", "CustomerName", "City", "Region"},
```

		SalesAmount	CustomerID.1	CustomerName	City	Region
1	1/4/2025	3000	C001	Customer 1	Legazpi	Luzon
2	2/12/2025	7500	C002	Customer 2	Manila	Luzon
3	3/11/2025	800	C003	Customer 3	Butuan	Mindanao
4	1/1/2025	45000	C004	Customer 4	Dumaguete	Visayas
5	3/3/2025	1200	C005	Customer 5	Dumaguete	Visayas
6	3/29/2025	2200	C006	Customer 6	Cebu	Visayas
7	2/7/2025	1200	C007	Customer 7	Zamboanga	Mindanao
8	1/17/2025	2200	C008	Customer 8	Baguio	Luzon
9	3/29/2025	45000	C009	Customer 9	Cagayan de Oro	Mindanao
10	2/19/2025	1200	C010	Customer 10	Cagayan de Oro	Mindanao
11	2/22/2025	3000	C011	Customer 11	Iloilo	Visayas
12	3/1/2025	3000	C012	Customer 12	Cagayan de Oro	Mindanao
13	3/7/2025	1200	C013	Customer 13	Cagayan de Oro	Mindanao
14	2/12/2025	45000	C014	Customer 14	Baguio	Luzon
15	1/6/2025	800	C015	Customer 15	Tacloban	Visayas
16	3/2/2025	45000	C016	Customer 16	Butuan	Mindanao
17	2/15/2025	45000	C017	Customer 17	Baguio	Luzon
18	2/10/2025	800	C018	Customer 18	Iloilo	Visayas
19	2/5/2025	7500	C019	Customer 19	Baguio	Luzon
20	1/28/2025	800	C020	Customer 20	Quezon City	Luzon
21	1/22/2025	800	C021	Customer 21	Manila	Luzon
22	1/14/2025	800	C022	Customer 22	Legazpi	Luzon
23	1/14/2025	3000	C023	Customer 23	Tacloban	Visayas
24	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Query Settings

PROPERTIES

Name

Orders

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Merged Queries
- Expanded Customers

Part 4: Validate Merged Data

- Check row counts.
- Verify no unexpected null values.
- Confirm matching customer details.

```
= Table.Combine({Orders, #"Orders (2)", #"Orders (3)"})
```

	order_id	customer_id	order_date	total_sales	customer_name	city
175	O119	C119	3/10/2025	45000	Customer 119	Iloilo
176	O120	C144	3/19/2025	2200	Customer 120	Dumaguete
177	O121	C076	3/23/2025	2200	Customer 121	Baguio
178	O123	C176	3/3/2025	7500	Customer 123	Cagayan de (
179	O130	C156	3/9/2025	45000	Customer 130	Cebu
180	O132	C093	3/9/2025	3000	Customer 132	Butuan
181	O141	C198	3/3/2025	3000	Customer 141	Dumaguete
182	O147	C128	3/1/2025	1500	Customer 147	Butuan
183	O152	C091	3/16/2025	1500	Customer 152	Baguio
184	O156	C152	3/3/2025	1200	Customer 156	Dumaguete
185	O159	C090	3/31/2025	3000	Customer 159	Tacloban
186	O169	C115	3/17/2025	1200	Customer 169	Dumaguete
187	O175	C145	3/1/2025	2200	Customer 175	Cebu
188	O180	C015	3/14/2025	1200	Customer 180	Dumaguete
189	O181	C047	3/27/2025	2200	Customer 181	Butuan
190	O184	C114	3/20/2025	3000	Customer 184	Cagayan de (
191	O185	C039	3/7/2025	45000	Customer 185	Butuan
192	O186	C155	3/7/2025	2200	Customer 186	Cagayan de (
193	O188	C009	3/18/2025	7500	Customer 188	Cagayan de (
194	O190	C153	3/4/2025	3000	Customer 190	Davao
195	O194	C198	3/30/2025	1500	Customer 194	Iloilo
196	O195	C012	3/16/2025	2200	Customer 195	Butuan
197	O200	C196	3/16/2025	3000	Customer 200	Zamboanga

Guide Questions

1. What is the difference between Merge and Append?

   - Merge combines tables side by side using a matching column (like a JOIN), while Append stacks tables on top of each other to add more rows.

2. Why use Inner Join instead of Left Join?

   - Inner Join only keeps records that match in both tables, giving cleaner and more accurate data. Left Join keeps all records from the left table even if there's no match, which can result in null values.

3. What happens to unmatched records in an Inner Join?

   - Unmatched records are completely removed — only rows with matching values in both tables are kept.

4. When is Append more appropriate than Merge?

   - Append is better when you have multiple tables with the same columns and just want to combine all their rows together, like monthly sales files.

Enhancement Activities

Try Left Join and compare results.

```
= Table.ExpandTableColumn(#"Merged Queries1", "Customers", {"CustomerID", "CustomerName", "City", "Region"},
```

	region	CustomerID	CustomerName	City.1	Region.1
1	Luzon	C001	Customer 1	Legazpi	Luzon
2	Visayas	C004	Customer 4	Dumaguete	Visayas
3	Luzon	C008	Customer 8	Baguio	Luzon
4	Visayas	C015	Customer 15	Tacloban	Visayas
5	Luzon	C020	Customer 20	Quezon City	Luzon
6	Luzon	C021	Customer 21	Manila	Luzon
7	Luzon	C022	Customer 22	Legazpi	Luzon
8	Visayas	C023	Customer 23	Tacloban	Visayas
9	Luzon	C027	Customer 27	Baguio	Luzon
10	Mindanao	C028	Customer 28	Cagayan de Oro	Mindanao
11	Luzon	C033	Customer 33	Baguio	Luzon
12	Visayas	C034	Customer 34	Iloilo	Visayas
13	Luzon	C035	Customer 35	Quezon City	Luzon
14	Luzon	C039	Customer 39	Quezon City	Luzon
15	Visayas	C040	Customer 40	Iloilo	Visayas
16	Visayas	C046	Customer 46	Tacloban	Visayas
17	Mindanao	C047	Customer 47	Zamboanga	Mindanao
18	Visayas	C048	Customer 48	Cebu	Visayas
19	Mindanao	C049	Customer 49	Zamboanga	Mindanao
20	Visayas	C050	Customer 50	Dumaguete	Visayas
21	Luzon	C051	Customer 51	Legazpi	Luzon
22	Luzon	C052	Customer 52	Quezon City	Luzon
23	Luzon	C053	Customer 53	Quezon City	Luzon
24	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Query Settings

PROPERTIES

Name

Orders

All Properties

APPLIED STEPS

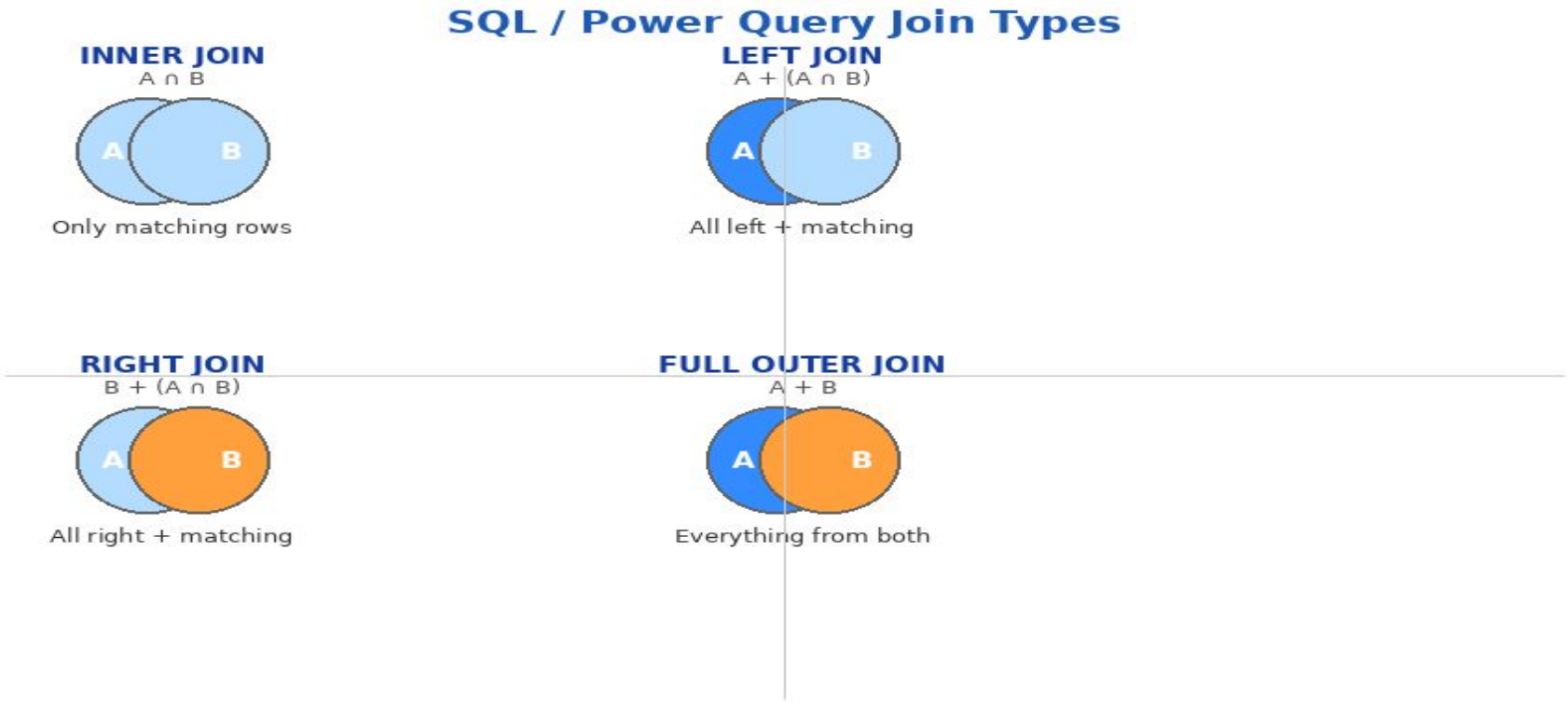
- Source
- Promoted Headers
- Changed Type
- Merged Queries
- Expanded Customers
- Removed Columns
- Renamed Columns
- Changed Type1
- Filtered Rows
- Merged Queries1
- Expanded Customers1

Identify unmatched records.

Queries [5]

- Orders
- Customers
- Orders (2)
- Orders (3)
- Combined_Sales

```
= Table.SelectRows(#"Expanded Customers", each ([CustomerID.1] = null))
```

		SalesAmount	CustomerID.1	CustomerName	City	Region
1	1/10/2025	45000	null	null	null	n
2	2/11/2025	45000	null	null	null	n
3	1/21/2025	2200	null	null	null	n
4	2/3/2025	1500	null	null	null	n
5	1/11/2025	3000	null	null	null	n
6	1/28/2025	800	null	null	null	n
7	1/6/2025	1200	null	null	null	n
8	2/27/2025	2200	null	null	null	n
9	3/5/2025	7500	null	null	null	n
10	3/17/2025	1200	null	null	null	n
11	1/26/2025	3000	null	null	null	n
12	3/18/2025	1200	null	null	null	n
13	1/17/2025	3000	null	null	null	n
14	1/17/2025	45000	null	null	null	n
15	1/5/2025	1500	null	null	null	n
16	2/26/2025	800	null	null	null	n
17	1/29/2025	1200	null	null	null	n
18	3/21/2025	1500	null	null	null	n
19	1/23/2025	3000	null	null	null	n
20	1/27/2025	1200	null	null	null	n
21	2/25/2025	45000	null	null	null	n
22	1/12/2025	7500	null	null	null	n
23	3/12/2025	1500	null	null	null	n
24	[illegible]	[illegible]	null	null	null	[illegible]

Create a diagram showing join types.